

Fricke-Self-Dual Eta-Quotients Are Eigenforms

A Corollary in Lean 4, and a Note on Its Motivation
in Persson–Volpato’s CHL S-Duality

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Status. This is a *preprint*. It has **not been peer reviewed**. The Lean development it describes compiles and is axiom-audited — those claims are machine-checked and independently reproducible — but the *exposition* has had no external referee. The mathematical content is a single closed corollary (Theorem 3.2) of an already-proved transformation law; the physics motivation cited in §5 is reported honestly as *motivation*, not as something this paper formalizes or proves. Read the scope before citing the result.

Abstract

We formalize, in Lean 4 on top of Mathlib, a corollary of an already-proved Fricke transformation law for eta-quotients: if the exponent vector r of $f(z) = \prod_{\delta|N} \eta(\delta z)^{r_\delta}$ satisfies $r_\delta = r_{N/\delta}$ for every $\delta \mid N$ (*Fricke-self-duality*), then f is a genuine eigenform of the Fricke transform $z \mapsto -1/(Nz)$, with a closed-form eigenvalue $\lambda = i^{-k} N^{k/2}$ depending only on the level N and weight k , not on r . We also prove exactly when $\lambda = \pm 1$. The only new mathematical content is a single product identity, $\prod_{\delta|N} \delta^{r_\delta} = N^k$ on self-dual r ; every other declaration is a direct substitution into the pre-existing transformation law. We report, as motivation and at the literature tier of this programme’s epistemic ladder — not as a formalized claim — that this self-duality condition is exactly Persson and Volpato’s “balanced Frame shape,” which they identify as characterizing CHL string compactifications self-dual under a Fricke-type S-duality on the heterotic axio-dilaton modulus. We state precisely where their normalization and ours diverge, so the two are not conflated.

MSC 2020: 11F11, 11F20, 68V20. **Keywords:** eta-quotients, Fricke involution, eigenforms, Lean 4, Mathlib, formal verification, CHL models.

1 Introduction

A companion development in this programme, *A Mathlib-Idiomatic Formalization of the Fricke Involution* [1], formalizes the matrix-level Fricke involution W_N on $\Gamma_0(N)$, and a second, *Eta-Quotients in Lean 4* [2], formalizes the transformation law of eta-quotients under W_N : for $f(z) = \prod_{\delta|N} \eta(\delta z)^{r_\delta}$ of weight k (i.e. $\sum_{\delta|N} r_\delta = 2k$),

$$f\left(\frac{-1}{Nz}\right) = i^{-k} N^k \left(\sqrt{\prod_{\delta|N} \delta^{r_\delta}}\right)^{-1} z^k f^*(z), \quad (1)$$

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where f^* is the *dual* quotient, exponent vector $\delta \mapsto r_{N/\delta}$. This is `etaQuotient_fricke` (tag F3.2-A7), already proved, sorry-free, in `EtaQuotientModularity.lean`. In general $f^* \neq f$: `fricke_dual_pin` in that module exhibits an r whose dual is a genuinely different function.

This note asks the natural next question: what happens when $f^* = f$? Equation (1) then collapses to a genuine eigenform relation, with an eigenvalue depending only on N and k . That collapse is short — a substitution and a simplification, not a fresh derivation — and its only nontrivial ingredient is one product identity (Lemma 2.2). We formalize it, together with a closed characterization of when the eigenvalue is exactly ± 1 , and we record, honestly and at arm’s length, why the question was worth asking: the self-duality condition on r is exactly the combinatorial shape of a condition in the physics literature that this paper does not attempt to derive.

2 The self-duality condition

Definition 2.1 (Fricke-self-dual exponent vector). *IsFrickeSelfDual*. For $N > 0$ and an exponent vector $r : \mathbb{N} \rightarrow \mathbb{Z}$,

$$\text{IsFrickeSelfDual}(N, r) :\iff \forall \delta \mid N, r_\delta = r_{N/\delta}.$$

We call this condition *self-dual* (equivalently *Fricke-symmetric*), never “balanced.” `EtaQuotientModularity.lean` already uses “balanced” for an unrelated concept — `exists_balanced_add`, tag F3.2-B1, the `Int.bmod` balanced-remainder-division step used in the $\Gamma_0(N)$ generator descent — and the physics literature discussed in §5 uses “balanced” for a third, related but distinct, sense. Reusing the word here would collide with both. `IsFrickeSelfDual` is `Decidable` (`decidableIsFrickeSelfDual`), so membership can be checked by `decide` on concrete instances, which every pin in the development below does.

Lemma 2.2 (The product collapses to a level power). *prod_zpow_selfDual*. For $N > 0$, r self-dual on N , and $\sum_{\delta \mid N} r_\delta = 2k$,

$$\prod_{\delta \mid N} \delta^{r_\delta} = N^k.$$

This is the only place the self-duality hypothesis does real work beyond a substitution: it forces the divisor-weighted product of the exponents to collapse to a single power of the level, which is what removes r from the eigenvalue entirely. It was checked, independently of the Lean proof, in exact rational arithmetic over four ranges (up to $1 \leq N \leq 60$, exponents in $[-4, 4]$, 671504 self-dual instances), zero mismatches, and against 9597 non-self-dual even-weight vectors of which 98.8% violate the identity — confirming self-duality is load-bearing for the collapse, not an artifact of the weight condition alone.

3 The eigenform relation

Definition 3.1 (Fricke eigenvalue). *frickeEigenvalue*. For $N > 0$ and $k \in \mathbb{Z}$,

$$\lambda(N, k) := i^{-k} \sqrt{N^k}.$$

Theorem 3.2 (Self-dual eta-quotients are Fricke eigenforms). *etaQuotient_fricke_selfDual*. For $N > 0$, r self-dual on N , $\sum_{\delta \mid N} r_\delta = 2k$, and z in the open upper half-plane,

$$f\left(\frac{-1}{Nz}\right) = \lambda(N, k) z^k f(z), \quad f(z) = \prod_{\delta \mid N} \eta(\delta z)^{r_\delta}.$$

The proof is two rewrites: substitute (1)’s dual quotient f^* by f itself (Definition 2.1 makes this a congruence over the divisor product, `etaQuotient_congr_divisors`), then collapse the constant using Lemma 2.2 (`fricke_const_selfDual`). Nothing else enters. An intermediate, unnormalized form retaining $\sqrt{\prod \delta^{r_\delta}}$ rather than λ is `etaQuotient_fricke_selfDual_raw`; a normalized form isolating the root of unity i^{-k} on one side is `etaQuotient_fricke_selfDual_normalized`.

Remark 3.3 (What this is not). Theorem 3.2 makes no reference to $\Gamma_0(N)$, to cusp conditions, or to the bundled type $M_k(\Gamma_0(N))$: it is a statement about one function value at one point, $-1/(Nz)$, transported from the already-proved general transformation law. Calling $\lambda(N, k)$ “the eigenvalue of the Fricke operator W_N on $M_k(\Gamma_0(N))$ ” would overstate what is proved here; the correct reading is that f is an eigenvector of the single linear map $g \mapsto g(-1/(Nz))$ restricted to functions of this shape.

3.1 When is the eigenvalue ± 1 ?

The eigenvalue’s dependence on N and k alone makes an exact characterization of $\lambda = \pm 1$ reachable without further hypotheses on r .

Theorem 3.4 (Modulus, and the sign criteria). *For $N > 0$ and $k \in \mathbb{Z}$:*

$$\begin{aligned} \|\lambda(N, k)\| &= \sqrt{N^k} && \text{(frickeEigenvalue_norm)} \\ \lambda(N, k) = 1 &\iff 4 \mid k \wedge (N = 1 \vee k = 0) && \text{(frickeEigenvalue_eq_one_iff)} \\ \lambda(N, k) = -1 &\iff k \bmod 4 = 2 \wedge N = 1 && \text{(frickeEigenvalue_eq_neg_one_iff)} \end{aligned}$$

Corollary 3.5 (± 1 in terms of the weight sum). *`frickeEigenvalue_pm_one_iff`. For $N > 0$, r self-dual, $\sum_{\delta|N} r_\delta = 2k$,*

$$\lambda(N, k) \in \{1, -1\} \iff 4 \mid \sum_{\delta|N} r_\delta \wedge (N = 1 \vee \sum_{\delta|N} r_\delta = 0).$$

Theorem 3.4 rests on two independent arithmetic facts, each proved and pinned on its own before being combined: the periodicity of $i^k \bmod 4$ (`I_zpow_emod`, genuinely new — Mathlib’s own `Complex.I_pow_eq_pow_mod` is stated only for $\mathbb{N}$ -exponents, not $\mathbb{Z}$), and $(N : \mathbb{R})^k = 1 \iff (N = 1 \vee k = 0)$ for $N > 0$ (`natCast_zpow_eq_one_iff`). Every pinned instance in both theorems is checked twice, by the general lemma and by an independent hand computation sharing no lemma with it; the informative instances are the ones where a naive guess would be wrong — e.g. $(N, k) = (6, 2)$, where $4 \mid k$ fails but $k \bmod 4 = 2$ holds and $\lambda \neq -1$ only because $N \neq 1$, the case that most directly tests whether the level condition was dropped by accident.

4 Verification status

The development is 333 declarations in 6437 lines, in one new module, `EtaQuotientFrickeSelfDual.lean`, imported into the library’s default build target. It is entirely sorry-free: 24 statement-graph nodes (two definitions, sixteen general lemmas and characterizations, five worked eigenvalue pins cross-checked against a q -product evaluation of η , and one statement-shape guard), all **proved**, none blocked, none open. 299 of the 333 declarations carry a build-failing `\#guard\msgs in \#print axioms` guard reporting exactly [`propext`, `Classical.choice`, `Quot.sound`]; the remainder are the pure data definitions (the pinned exponent vectors) and auxiliary lemmas that a guard would not strengthen. **lake build SocrateAI** — the whole library’s shared default target, now also carrying this module’s guards — completes in 3769 jobs with no errors.

Every headline theorem was checked, independently of its Lean proof, against a direct q -product evaluation of the Dedekind eta function (a 600-term truncation, three points

per instance, relative error $\leq 1.8 \times 10^{-14}$), and Theorem 3.2's $N = 1$, $r \equiv 24$, $k = 12$ instance reproduces `eta_S_via_fricke`, itself independently re-derived from Mathlib's own `discriminant_S_invariant`. Kernel proof terms were inspected directly (`\#print`) to confirm the headline theorems are genuine substitutions into `etaQuotient_fricke` rather than restatements that merely typecheck: each reduces to at most two `rw` steps plus `ring`.

The statement-dependency graph (`dag/theorems.jsonl`) validates: 104 nodes, 95 proved, acyclic. No axiom beyond Lean's three standard axioms and the library's six pre-existing, disclosed `physics\_postulate\_$_i$` placeholders (unused by anything proved here) is declared anywhere in the library.

5 Motivation: Persson–Volpato's Fricke S-duality in CHL models

Nothing in this section is formalized. Nothing in §§2.1–4 names or depends on any object introduced here. We report the motivation for asking the question in the first place, at the literature (L) tier of this programme's epistemic ladder, because the disclosure norm this programme follows [7, 8] requires attributing sources plainly rather than letting a formal artifact imply more provenance than it has.

Persson and Volpato [4] study CHL compactifications — orbifolds of the heterotic string (equivalently, by string duality, of type IIA on $K3 \times T^2$) by an order- N symmetry g — and attach to g a *generalized Frame shape* $g \leftrightarrow \prod_{a|N} a^{m(a)}$, an exponent vector on the divisors of N , constrained by $\sum_{a|N} m(a) \cdot a = 24$ (their eq. 1.4, recording a 24-dimensional representation). They call the Frame shape *balanced* (their eq. 1.6) when $m(N/a) = m(a)$ for all $a \mid N$ — exactly Definition 2.1, read on their exponent vector — and their headline result (their §1.1, §3) is that a CHL model is self-dual under a Fricke S-duality $S \mapsto -1/(NS)$ on the heterotic axio-dilaton modulus S precisely when its Frame shape is balanced. That identification is established by physical arguments this Lean library does not contain and this note does not attempt: charge-lattice N -modularity, Conway-group representation theory on the 24-dimensional representation, and Witten-index/BPS-state counting.

Remark 5.1 (Two normalizations that must not be merged). Persson–Volpato's weight constraint $\sum_{a|N} m(a) \cdot a = 24$ is *not* the eta-quotient weight normalization $\sum_{\delta|N} r_\delta = 2k$ used throughout this paper. The two agree only at $N = 1$. Their canonical balanced shape at $N = 2$, $1^8 2^8$, satisfies $\sum a \cdot m(a) = 8 + 16 = 24$ but $\sum m(a) = 16$, i.e. weight 8, not 12 (checked this session over $1^k N^k$, $k = 24/(1 + N)$, for $N \in \{1, 2, 3, 5, 7, 11, 23\}$). A reader translating between the two literatures must carry the exponent vector's actual weight, not assume weight 12 throughout.

The modulus S in Persson–Volpato is the heterotic axio-dilaton, not the T^2 complex-structure modulus τ that a companion negative result in this programme [1] investigated and found *not* supported, at level $N > 1$, by the natural T-duality reference (Giveon–Porrati–Rabinovici). That the same Fricke matrix $W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$ recurs attached to a *different* modulus in a correctly-sourced reference, while failing to attach to the T^2 modulus in an incorrectly-sourced one, is exactly the distinction that negative result was designed to surface, and this note's citation discipline is a direct continuation of it: cite the source that actually supports the specific claim being motivated, not the source that merely mentions a similar-looking matrix.

6 Related work

Martin [3] gives the classical account of Fricke and Atkin–Lehner eigenvalue multiplicativity for eta-quotients; Theorem 3.2 is the special case of that classical picture reachable directly

from the already-proved general transformation law, stated and checked in Lean rather than re-derived. `etaQuotient_fricke` (§1)[2] supplies the general law this note specializes; [1] supplies the matrix-level Fricke involution W_N that both eta-quotient papers’ transformation laws are ultimately stated relative to, and its own T-duality negative result is the direct precedent for this note’s citation discipline in §5.

7 Limitations

Remark 7.1 (What is not done here).

- (a) No connection to $\Gamma_0(N)$ -modularity, cusp behaviour, or the bundled type $M_k(\Gamma_0(N))$ is made (Remark 3.3).
- (b) The physics claim motivating this note (§5) is not formalized, is not attempted here, and should not be inferred from the Lean content.
- (c) $\lambda(0, k) = 0$ is junk (the definition is total but the hypothesis $N > 0$ is required by every theorem that uses it); no theorem here relies on or claims anything about $N = 0$.
- (d) The characterization of $\lambda = \pm 1$ (Theorem 3.4) is complete for the eigenvalue itself, but Corollary 3.5 inherits the self-duality *hypothesis* on r — it does not characterize which exponent vectors are self-dual, a separate combinatorial question this note does not address beyond Definition 2.1’s decidability.

8 Conclusion

We have formalized, in Lean 4, that a Fricke-self-dual eta-quotient is an eigenform of the Fricke transform, with a closed-form eigenvalue depending only on level and weight, and exactly when that eigenvalue is ± 1 . The mathematics is a short, honest corollary of an already-proved transformation law; its interest is less in the difficulty of the proof than in the fact that the same combinatorial condition organizes a physical duality in the CHL string-compactification literature, reported here strictly as motivation, with the two literatures’ differing normalizations stated explicitly so they cannot be silently conflated.

A note on method: AI assistance and human responsibility

This note follows the disclosure norm argued for by Tao [7], endorsing recommendations of the Leiden Declaration on Artificial Intelligence and Mathematics [8]: disclose tool use, affirm that credit and responsibility belong to the human author, and attribute sources where AI tools cannot reliably do so themselves.

What AI did. The Lean 4 formalization, the literature search, and the drafting of this text were produced with the assistance of large language models (Anthropic’s Claude family), working under human direction across multiple sessions. The methodology is neuro-symbolic in the specific sense Tao describes [7]: natural-language reasoning proposes definitions, proof strategies, and prose, and the Lean kernel — not the model, and not the author’s trust in the model — is what admits or rejects every mathematical claim. Every theorem in this paper compiles against Mathlib with no `sorry` and a build-failing axiom guard on its footprint (§4); nothing here is asserted true because a model said so.

What the author did. Xavier Callens set the research direction — choosing this specialization specifically because it was reachable from already-proved machinery rather than requiring a fresh physics-bridge attempt, after a companion run in this programme found such an attempt correctly refused [1] — judged what the result means and whether it is worth publishing, and

is responsible for every claim in this document, including its errors. The physics/mathematics boundary in §5 was treated as a hard constraint on the formalization itself, not merely on the prose: no Lean declaration in this development names or depends on any physical object, verified by direct inspection and independently re-verified by an adversarial review pass. Tao’s own test is apt: a result that cannot be explained and defended by its author, formally verified or not, is not ready to be called complete [7]. The mathematical content here — why self-duality collapses the product, and why the two normalizations in §5 diverge — is something the author can explain without further AI assistance; the Lean proof terms that certify it were substantially AI-drafted and are trusted only because the kernel, not the author, checks them.

References

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